SonicVitals

Dashboard graphs: what they show, how they are calculated, and which methods and modes are compared

Contactless heart rate and breathing from a camera (Mac webcam or ESP32-S3 + GC2145) with an INMP441 microphone channel. This guide follows the code in *backend/* and *Frontend/* as of 17 October 2026.

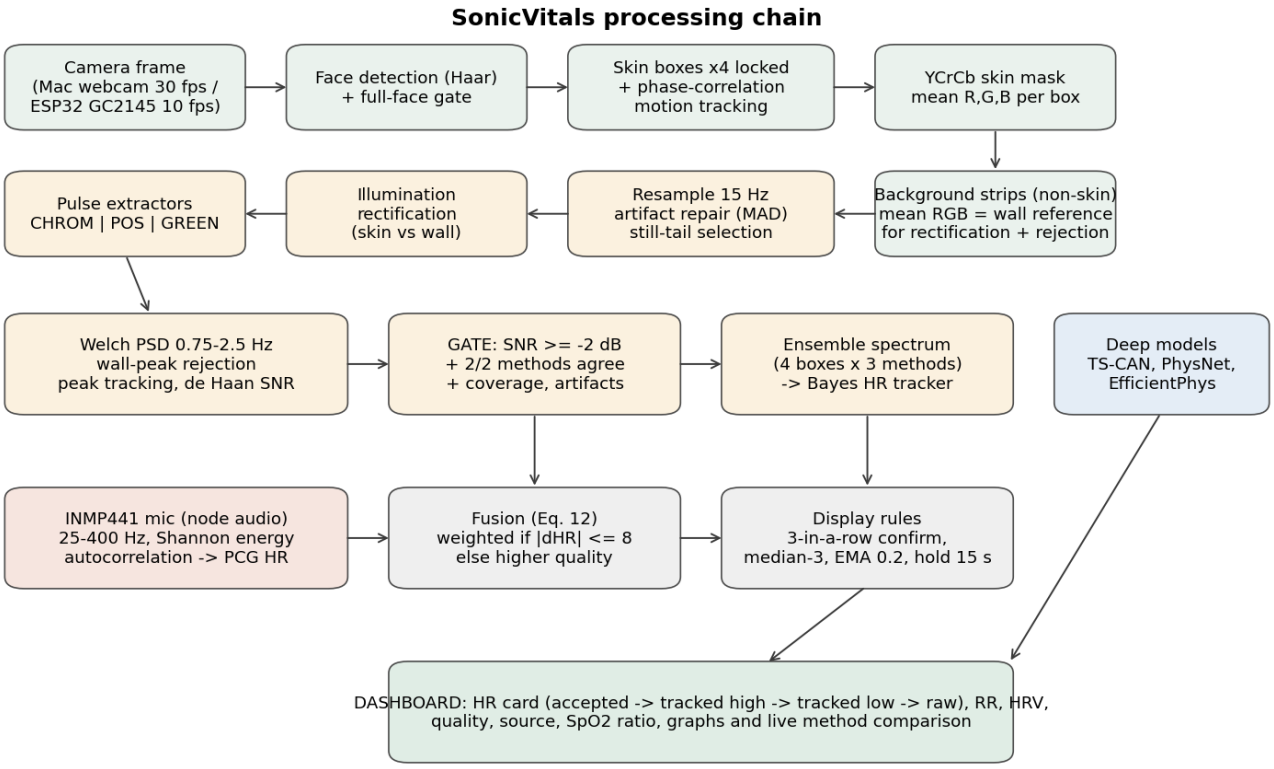


Figure 1. Processing chain. Green: image stage; amber: signal stage; blue: deep models (comparison only); red: microphone; grey: fusion and display.

About the example figures. The plots in this document are the dashboard's own telemetry, produced by replaying a 45 s webcam recording (16 Oct 2026) through the live engine. That recording has no reference pulse and most of its windows were *rejected* by the quality gate, so the figures show what the graphs look like and how a rejection appears, not a validated measurement.

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1. The processing chain, step by step

Every graph is a view of one stage of this chain. The camera path runs on every frame; the heart-rate estimate is recomputed every 2 s on the last 30 s of gated frames.

#	Stage	What happens	Key settings
1	Capture	Mac webcam frames (about 30 fps) or the ESP32-S3 node. The node computes skin-box colour sums on-board at 10 fps and streams JPEG video at 5 fps over Wi-Fi.	Node exposure is metered on the skin (target level 165).
2	Face detection and gate	Haar cascade on a copy at most 320 px wide; box smoothed. The full-face gate needs a frontal face of at least 20 % of frame width, fully inside the frame. Only gated frames enter the analysis buffer.	Box jitter under 6 % of the face size is ignored.
3	Skin boxes	Four boxes relative to the face: forehead, left cheek, right cheek, nose. The boxes are locked and follow the face with sub-pixel phase correlation against a template taken at lock time, so the same skin stays in each box when the head drifts.	Re-lock when tracking fails for 8 frames or the face moves more than 25 % of its width.
4	Skin mask and means	A YCrCb skin mask per box (computed once per lock) removes hair, brows and beard; the mean R, G, B of the remaining pixels gives one sample per box per frame.	Cr 133-180, Cb 77-135.
5	Background reference	Mean colour of non-skin strips beside and above the face: it sees the room lighting but has no pulse.	Used only when it carries a real lighting change (>= 30 % of skin band RMS).
6	Resampling and repair	Interpolate to a uniform 15 Hz grid; bridge gaps up to 2 s. Frame-to-frame jumps larger than 4 robust SDs (motion, re-lock) are zeroed in the log-derivative and re-integrated. After a movement only the still stretch since then is analysed (if at least 12 s).	Segment >= 10 s for HR, >= 15 s for RR; coverage >= 80 %.
7	Illumination rectification	Per colour channel, the skin trace is regressed on the background in 0.6-3 Hz and the shared part (flicker, auto-exposure) is subtracted.	beta 0-1.5.
8	Pulse extraction	Three classic projections of RGB to a pulse: CHROM (primary), POS and GREEN.	CHROM windows of 3 s; POS windows of 1.6 s.
9	Spectrum and peak	Welch power spectrum of the drift-removed pulse; the strongest peak in 0.75-2.5 Hz (45-150 BPM), staying near the last accepted HR when that peak is at least half as strong. Peaks equally strong on the background are skipped as lighting.	NFFT 4096; 0.6 Hz high-pass.
10	Quality gate	Accept only if de Haan SNR >= -2 dB, POS and GREEN both find the CHROM peak within 6 BPM, coverage >= 80 % and at most 15 % of frames repaired.	Thresholds set on synthetic data and own recordings, never on the test datasets.
11	Ensemble + tracker	Spectra of 4 boxes x 3 extractors are normalised and median-fused; a Bayes filter on a 0.5 BPM grid tracks HR over time and reports a confidence (posterior mass within +-3 BPM).	High confidence >= 0.8.
12	Microphone	INMP441 at the node rate (16 kHz, or 4 kHz in older recordings): 25-400 Hz band-pass, Shannon-energy envelope, autocorrelation period gives PCG HR; S1 peak times give RMSSD.	Envelope peak rho >= 0.4, beat consistency >= 0.6.
13	Fusion and display	Quality-weighted fusion of camera and mic when within 8 BPM, else the higher-quality one. A new HR is shown after 3 accepted values within +-5 BPM in 12 s; median of 3, EMA 0.2; held at most 15 s.	

2. Dashboard tab graphs and cards

2.1 Primary vision stream (video with skin boxes)

What it shows	Live camera image with the face box and the four skin boxes (green when the full-face gate passes). A badge says FULL FACE (GATE PASSED), an alignment hint, or SEARCHING.
How it is computed	Stages 2-3 above. The boxes are drawn at their motion-compensated position, so they should stay on the same patch of forehead, cheeks and nose while you move slightly.
How to read it	If the boxes slide onto hair, eyebrows or background, the signal will be poor. A camera-switch icon toggles Mac webcam / ESP32 camera.

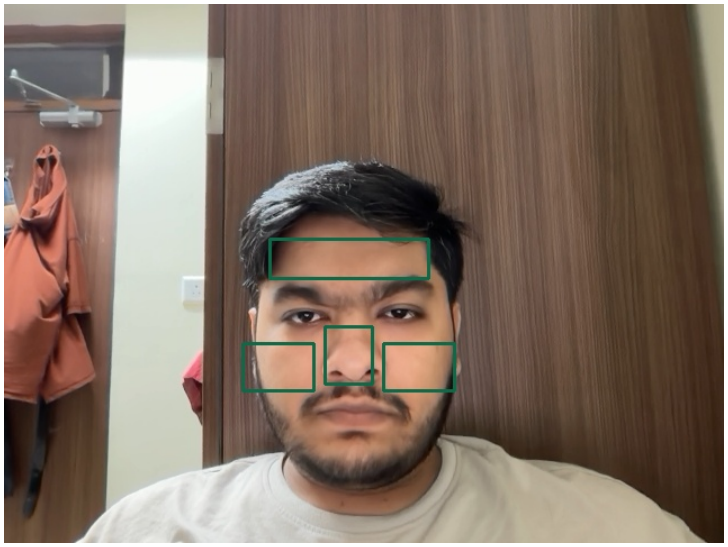


Figure 2. Skin boxes on a webcam frame.

2.2 Readout cards

Card	Value shown	How it is computed / when it is empty
HEART RATE	Accepted fused HR; otherwise tracked HR (high confidence); otherwise tracked HR (low confidence, amber); otherwise the raw camera peak (amber); otherwise mic-only estimate.	The caption states which of these is shown, plus SNR or confidence and the reason for rejection (e.g. SNR BELOW GATE, METHODS DISAGREE 1/2, MOVEMENT).
RESPIRATORY RATE	Breaths/min from the green trace.	Peak of the Welch spectrum in 0.15-0.40 Hz on a >= 15 s segment; accepted if >= 50 % of 0.05-1 Hz power lies at the peak, else shown amber.
HRV (RMSSD)	ms	From heart-sound (PCG) beat times when >= 20 clean intervals exist; otherwise an exploratory camera RMSSD from pulse peaks when the track is high confidence.
SIGNAL QUALITY	%	When accepted: 100 x max(camera weight, mic rho). Camera weight = spectral concentration (share of in-band power within +-0.1 Hz of the peak) x gate. Otherwise the camera concentration in amber with the SNR and gate.
HR SOURCE	FUSED / OPTICAL / ACOUSTIC / CAMERA (TRACKED)	Which channel produced the accepted HR; CONFLICT when camera and mic differ by more than 8 BPM.
SpO2 (EXPLORATORY)	--	Only the ratio-of-ratios R = (AC/DC red)/(AC/DC blue) is computed. A number needs calibration against a finger oximeter, so none is shown.

2.3 Microphone waveform (PCG)

What it shows	The last 4 s of the INMP441 signal from the ESP32 node.
How it is computed	Node audio (16 kHz; 4 kHz in older recordings), 25-400 Hz Butterworth band-pass (heart-sound band), decimated to 1 kHz, min/max decimated to 600 points and normalised to +-1.
How to read it	With the mic on the chest, S1/S2 heart sounds appear as regular bursts. Off the chest it is noise, and the mic HR is withheld by the beat-consistency check.

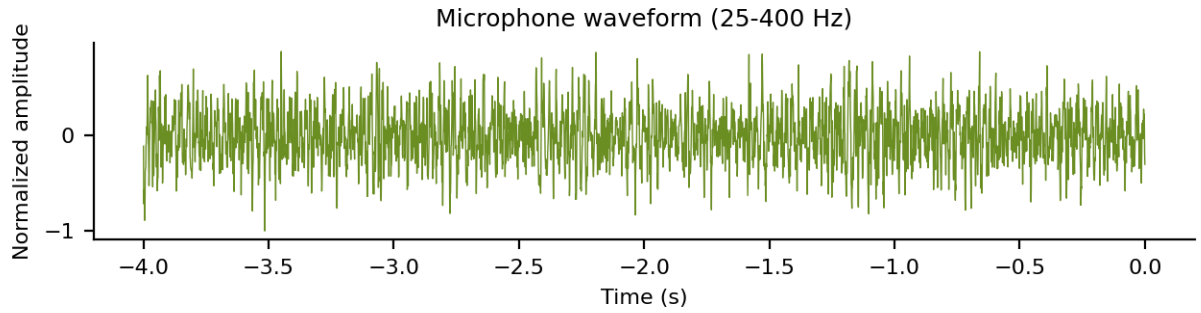


Figure 2b. Node microphone, 4 s (14 Oct recording): noise, no heart sounds.

2.4 Recovered pulse (CHROM, 0.75-2.5 Hz) and respiratory waveform

What it shows	Top: the blood-volume pulse recovered from skin colour. Bottom: the slow breathing oscillation.
How it is computed	Pulse: CHROM projection of the 15 Hz skin RGB, band-passed 0.75-2.5 Hz (3rd-order Butterworth, zero phase), z-scored. Respiration: green channel band-passed 0.15-0.40 Hz, z-scored.
How to read it	A good pulse looks like a regular wave at your heart rate (about 1 cycle per second at 60 BPM). Irregular amplitude or several frequencies mixed means noise or motion dominate.

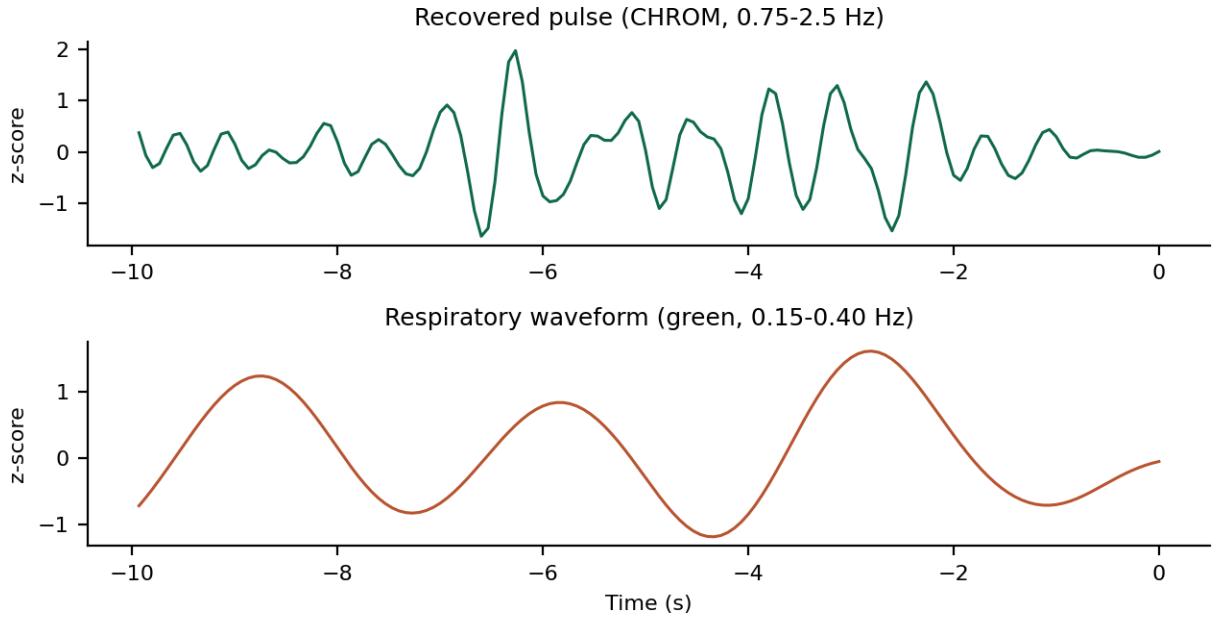


Figure 3. Pulse and respiration traces (replayed clip).

2.5 Heart-rate history (last 5 min)

What it shows	Every second: the displayed HR (solid line), each camera estimate as an accepted (filled) or rejected (hollow) point, microphone HR (diamonds) and the tracked HR (dotted: green = high confidence, amber = low).
How it is computed	Points come from the 2 s estimates (stage 9-10); the tracked line from stage 11. Hovering a rejected point shows why it was rejected.
How to read it	Accepted points and the green tracked line should sit on one level when you are still. Hollow points scattered widely mean the camera signal is not usable at that moment.

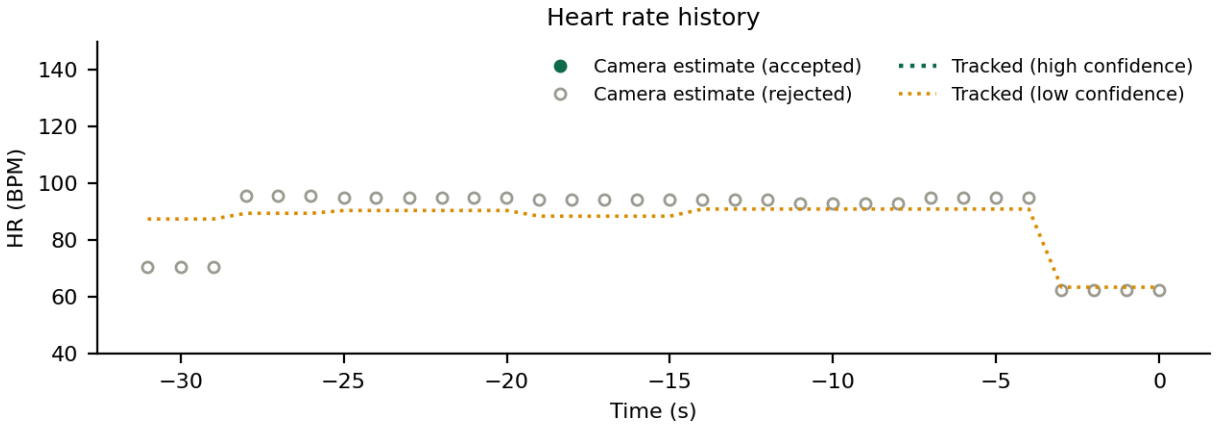


Figure 4. History on the replayed clip: windows rejected (hollow); tracker below high confidence (amber).

2.6 Signal quality vs acceptance gate

What it shows	Pulse SNR over time (black), the gate threshold (dashed, -2 dB) and, on the right axis, how many of POS and GREEN agree with the CHROM peak (0, 1 or 2).
How it is computed	SNR (de Haan & Jeanne 2013): power within +-0.1 Hz of the HR peak and +-0.2 Hz of its harmonic, divided by the rest of 0.5-4 Hz, in dB.
How to read it	An estimate is accepted only when the black line is above the dashed one *and* the amber step is at 2. On the replayed clip SNR stays below the gate, so nothing is accepted.

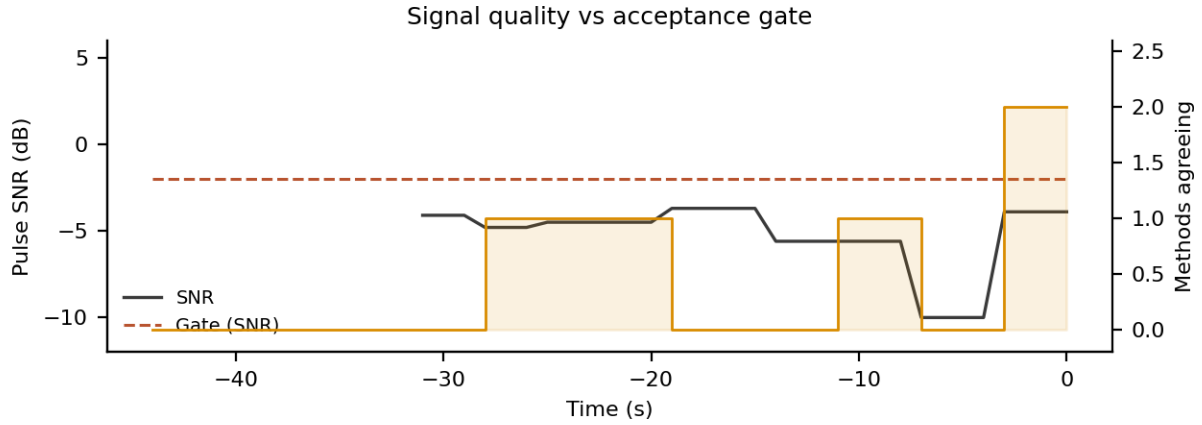


Figure 5. Quality against the gate (replayed clip).

2.7 Method comparison (live, same face, same moment)

What it shows	A table and a dot chart with the heart rate from every method at the same moment, the median (dotted line) and the spread; AGREE when the spread is <= 6 BPM.
How it is computed	SonicVitals, CHROM, POS, GREEN from the skin boxes every 2 s; TS-CAN, PhysNet, EfficientPhys on the last 30 s of 72 x 72 face crops every 12 s (needs >= 12 fps, so Mac webcam only); microphone from the node. Section 6 describes each method.
How to read it	Methods that share the same input but different algorithms should agree when the signal is good. A wide spread is itself a sign of a poor signal.

3. Signal Analysis dashboard (laboratory graph view)

The Signal Analysis tab is the engineering view of the camera pulse: one strip of six live readouts, six graphs in a 3 x 2 grid, and a CSV export button. Everything on it is refreshed from */api/vitals* about every 0.5 s while the tab is open; the numbers come from the most recent 2 s estimate on the last 30 s of gated frames (stages 6-10 of the chain).

3.1 Layout

Position	Element	Section
Header	Title "SIGNAL ANALYSIS PIPELINE" and the EXPORT RECORDING (CSV) button	3.9
Top strip	Six readouts: cardiac peak, SNR, spectral concentration, respiration, methods agreeing | motion, video source and fps	3.2
Row 1	Left: rPPG power spectral density. Right: chest microphone spectrogram	3.3, 3.4
Row 2	Left: standardized pulse signal. Right: moving-average decomposition	3.5, 3.6
Row 3	Left: skin colour change (R, G, B). Right: method agreement CHROM vs POS vs green	3.7, 3.8

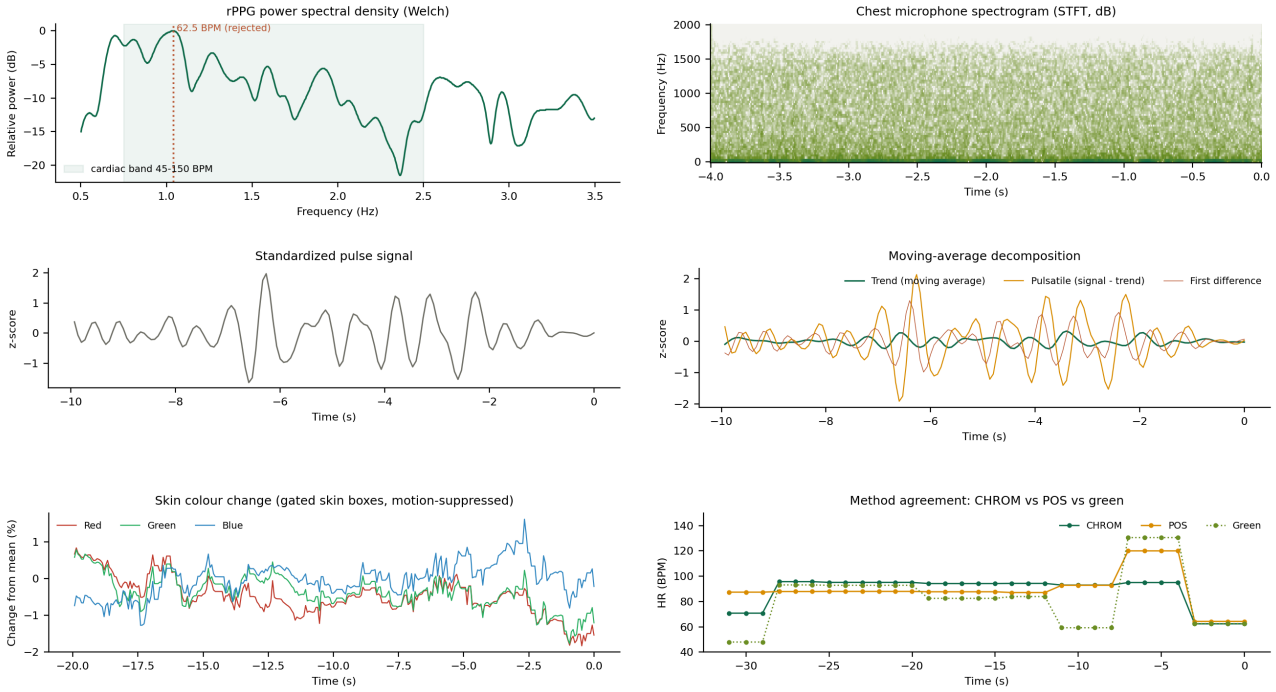


Figure 6. The six Signal Analysis graphs in the tab's grid order (camera graphs: replayed webcam clip, 16 Oct; spectrogram: node microphone recording, 14 Oct).

3.2 Telemetry strip (the six readouts)

Values in the third column are what the strip showed at the end of the replayed clip. Its estimate was **rejected** by the gate.

Readout	How it is computed	Example	How to read it
CARDIAC PEAK (f0)	Frequency of the chosen peak of the CHROM spectrum in 0.75-2.5 Hz (with peak tracking and lighting peaks skipped); BPM = 60 x f0. Shows "NEEDS 10 s SEGMENT" until 10 s of gated frames exist.	1.04 Hz (62.5 BPM)	The candidate heart rate before the gate; it is the dotted line in the spectrum.
SNR (DE HAAN)	10 log10 of power at f0 (+-0.1 Hz) and 2 f0 (+-0.2 Hz) over the rest of 0.5-4 Hz.	-3.9 dB	Accepted only at >= -2 dB. Around -5 dB the peak is barely above noise.
SPECTRAL CONCENTRATION (Q)	Share of the 0.75-2.5 Hz power within +-0.1 Hz of f0, in %.	33.2%	Higher = one dominant peak; a flat spectrum gives a few %. It is also the camera weight in fusion.
RESPIRATION (fr)	Peak of the green-channel spectrum in 0.15-0.40 Hz on >= 15 s; shown only if >= 50 % of 0.05-1 Hz power lies at that peak.	-- Hz (-- BrPM) (best guess 17.4 BrPM, below the 50 % rule)	Breathing frequency in Hz and breaths/min.
METHODS AGREEING | MOTION	How many of POS and GREEN find the CHROM peak within 6 BPM (0-2); share of frame pairs repaired by the artifact filter; seconds skipped before the latest still stretch.	2/2 | 14% REPAIRED	Needs 2/2 and <= 15 % repaired to be accepted.

Readout	How it is computed	Example	How to read it
VIDEO SOURCE (MEASURED FPS)	HOST WEBCAM or ESP32-S3 CAMERA, and the frame rate actually processed.	Mac webcam about 30 fps	Below about 10 fps the pulse is poorly sampled; deep models need >= 12 fps.

3.3 rPPG power spectral density (Welch) [row 1, left]

What it shows	Power of the pulse signal against frequency, with the cardiac band shaded (0.75-2.5 Hz) and the chosen peak marked (green = accepted, red = rejected) with its BPM.
How it is computed	Welch periodogram (256-sample segments, NFFT 4096) of the 0.6 Hz high-passed CHROM pulse at 15 Hz, in dB relative to the maximum, shown 0.5-3.5 Hz.
How to read it	A clean recording has one sharp peak (and a smaller one at twice the frequency). Many peaks of similar height, as here, mean the window holds no dominant rhythm and the gate rejects it.

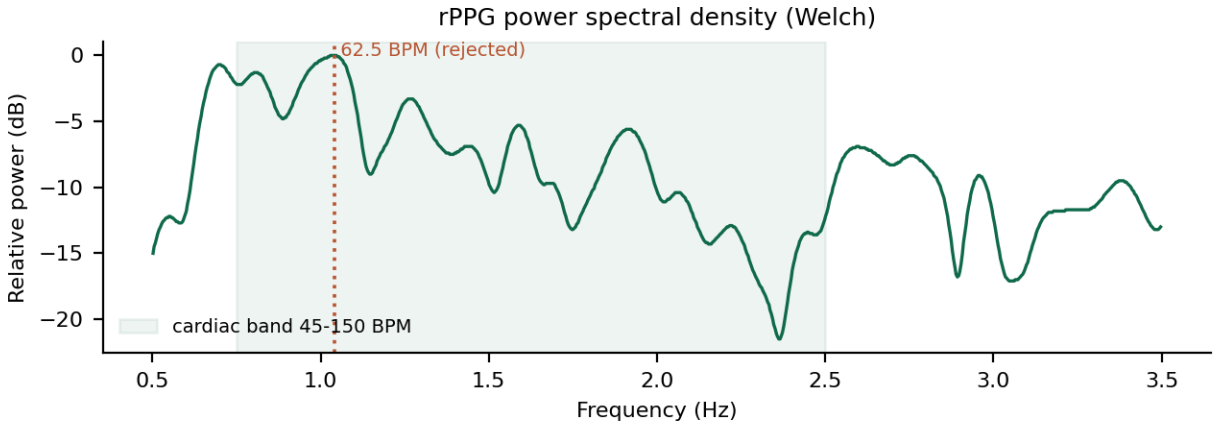


Figure 7. Spectrum of the replayed clip: several peaks of similar height.

3.4 Chest microphone spectrogram (STFT, dB) [row 1, right]

What it shows	Time-frequency picture of the last 4 s of node audio.
How it is computed	Short-time Fourier transform with a Hamming window of about 32 ms (512 samples at 16 kHz, 128 at 4 kHz), 75 % overlap, up to 2 kHz (or the Nyquist frequency), averaged into 60 time columns, dB relative to the maximum, colour range -60 to 0 dB.
How to read it	Heart sounds appear as short vertical stripes below about 200 Hz repeating at the heart rate. Broadband speckle across all frequencies, as in this recording, is background noise: no heart sounds are visible, which is what happens when the mic is not pressed on the chest.

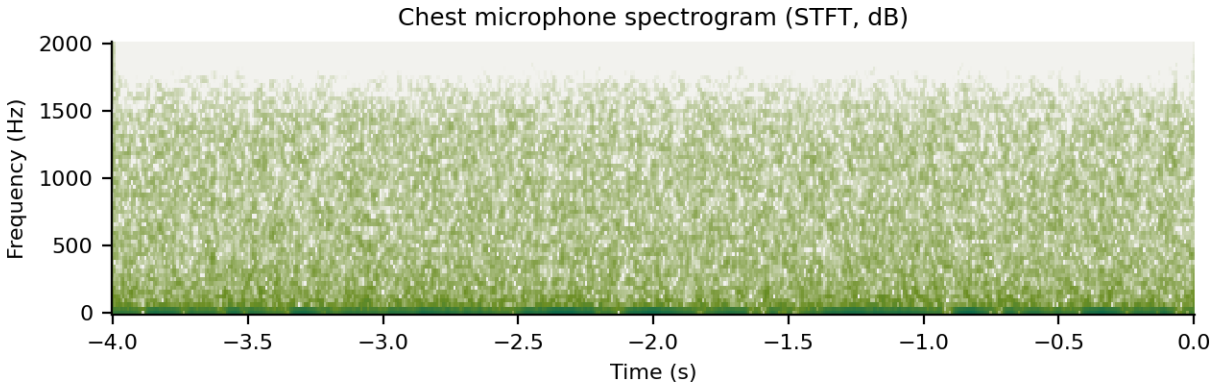


Figure 8. Node microphone, 4 s from the 14 Oct recording (4 kHz).

3.5 Standardized pulse signal [row 2, left]

What it shows	The recovered blood-volume pulse, larger than on the dashboard, for inspecting its shape.
How it is computed	CHROM pulse band-passed 0.75-2.5 Hz (3rd-order Butterworth, zero phase) and z-scored (mean 0, SD 1); last part of the analysis buffer at 15 Hz.
How to read it	Look for a regular wave with one cycle per heartbeat and steady amplitude. Bursts of large amplitude (here around -6 s) are usually motion or lighting, not the heart.

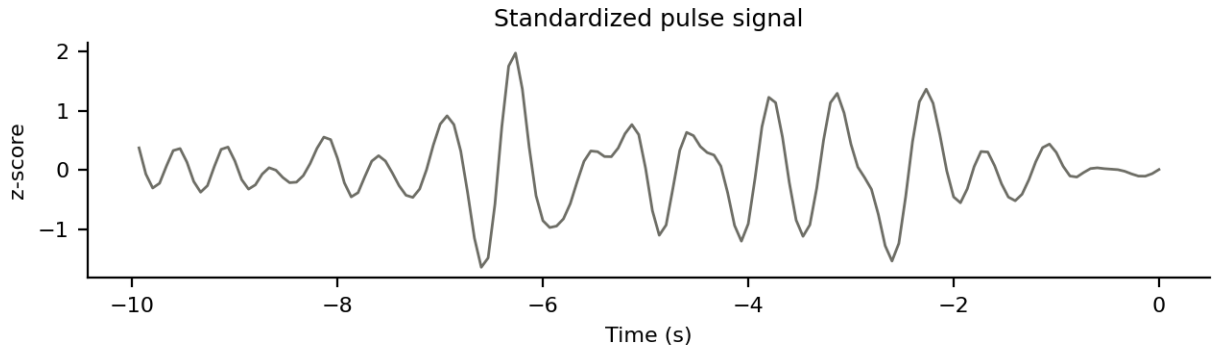


Figure 9. Standardized pulse (replayed clip).

3.6 Moving-average decomposition [row 2, right]

What it shows	Splits the pulse into a slow trend, the pulsatile part (signal minus trend) and the first difference.
How it is computed	Trend = moving average over W = min(14, N/4) samples (about 1 s at 15 Hz); pulsatile = signal - trend; first difference = x[n] - x[n-1]. The pulsatile/trend energy ratio is reported as motion_ratio.
How to read it	A large trend means slow drift (lighting, posture); spikes in the first difference mark sudden changes such as motion.

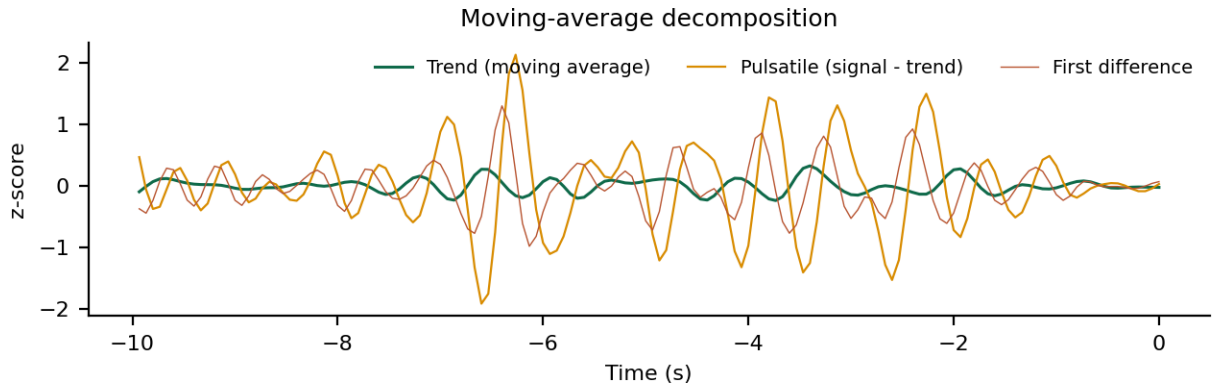


Figure 10. Decomposition (replayed clip).

3.7 Skin colour change (gated skin boxes, motion-suppressed) [row 3, left]

What it shows	The raw input of the whole chain: percentage change of the mean red, green and blue of the skin boxes over the last 20 s.
How it is computed	Box means (skin-masked, motion-compensated) after artifact repair and illumination rectification, divided by their mean, minus 1, x 100.
How to read it	The pulse is only about 0.3 % (mostly in green). The +-1-2 % drifts seen here come from head pose, shading and webcam auto-exposure; they are what pushes SNR below the gate.

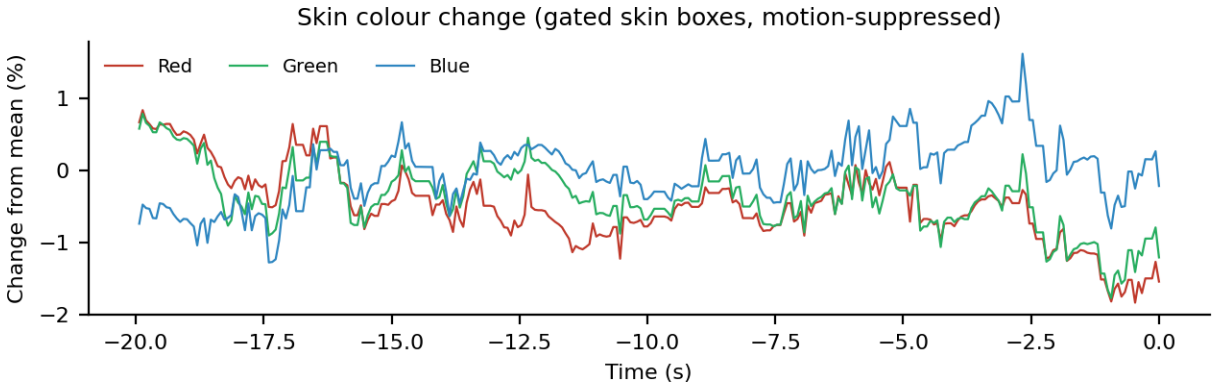


Figure 11. Skin colour change (replayed clip).

3.8 Method agreement over time (CHROM vs POS vs green) [row 3, right]

What it shows	Heart rate from each of the three extractors at every update over the last 5 min.
How it is computed	Each extractor's pulse goes through the same spectrum and peak search (stage 9).
How to read it	The lines overlap when the pulse is real; noise gives method-specific peaks, which is why agreement is part of the gate. GREEN is the weakest method (it has no motion or lighting rejection).

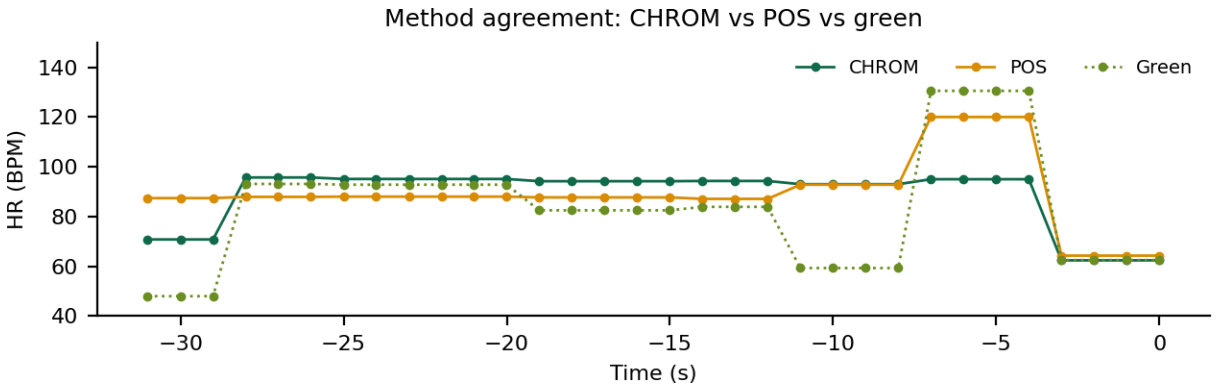


Figure 12. Extractor agreement (replayed clip).

3.9 EXPORT RECORDING (CSV)

Downloads the display log kept by the browser (one row per display update) with columns: *time_iso, hr_display_bpm, rr_display_brpm, rmssd_ms, hr_source, hr_optical_bpm, hr_acoustic_bpm, quality_pct, video_source*. Raw frames, colour traces and audio are saved separately by the host session recorder (*data/sessions/<id>/*: frames.csv, node_rgb.csv, estimates.csv, audio.wav).

4. Sessions tab (optical vs acoustic agreement)

A session records the camera HR and the microphone HR together for a fixed time (default 60 s) and compares them. Neither is a medical reference, so this measures *agreement*, not accuracy.

4.1 Bland-Altman plot

What it shows	For every paired update: x = mean of optical and acoustic HR, y = optical minus acoustic. Solid line = mean difference (bias); dashed lines = limits of agreement.
How it is computed	Bias = mean(d); SD = standard deviation of d; limits = bias +- 1.96 SD. Also listed: MAE, RMSE and Pearson r of the pairs.
How to read it	Points close to 0 and narrow limits mean the two sensors agree; a trend of y with x means one channel over- or under-reads at high rates.

4.2 Optical and acoustic HR trend

What it shows	Both heart rates against elapsed session time (2 s updates).
How it is computed	Optical = accepted camera HR; acoustic = PCG HR (stage 12).
How to read it	The two lines should move together.

5. Diagnostics Studio (colour indices)

On CAPTURE SNAPSHOT, the sharpest recent frame with a full face is analysed in five regions relative to the face box (forehead, both cheeks, eye band, lips). The indices are computed from 8-bit camera values, are **uncalibrated**, and their ranges are software ranges, not clinical thresholds.

Index	Formula (region means)	Meaning
Erythema EI	100 x (log10 R - log10 G), forehead	Redness of the skin.
Pallor PI	(G + B of both cheeks) / (2 x (R left + R right))	Paleness: less red relative to green/blue.
Eye-band yellow YI	100 x (R + G - 2B) / (R + G + B), eye band	Yellowness around the eyes.
Lip B/R CR	B / R, lips	Blueness of the lips relative to red.
Facial asymmetry HFI	100 x |G left - G right| / (G left + G right)	Left/right colour difference (also flags uneven lighting).

6. Modes and methods compared

6.1 Heart-rate card modes (what the big number is)

Mode	Condition	Shown as
Accepted	Passed the full gate (SNR, 2/2 agreement, coverage, artifacts) and the display confirmation (3 values within +-5 BPM in 12 s).	Black value, "ACCEPTED", SNR.
Tracked, high confidence	No accepted value, but the Bayes tracker holds >= 80 % of its probability within +-3 BPM. On simulations pure noise never reached this level.	Black value, "TRACKED, HIGH CONFIDENCE xx %".
Tracked, low confidence	Tracker has a best guess below 80 % confidence.	Amber value with the reason.
Raw camera	Only the single-window CHROM peak exists.	Amber value with the rejection reason.
Microphone only	No face; the mic has an estimate but irregular beats.	Amber, "MICROPHONE ONLY".

6.2 Methods in the live comparison

Method	Family	Input	Principle	Runs
SonicVitals	Ours	4 skin boxes, background	CHROM + POS + GREEN with artifact repair, illumination rectification, wall-peak rejection, SNR + consensus gate; ensemble spectrum + Bayes tracker for the continuous estimate.	Every 2 s, about 30 ms
CHROM	Signal processing	Skin boxes	de Haan & Jeanne 2013: two chrominance signals X = 3R - 2G, Y = 1.5R + G - 1.5B; pulse = X - alpha Y with alpha = std X / std Y (cancels specular light and motion).	Every 2 s
POS	Signal processing	Skin boxes	Wang et al. 2017: projection on the plane orthogonal to skin tone: S1 = G - B, S2 = -2R + G + B; pulse = S1 + (std S1 / std S2) S2, 1.6 s windows.	Every 2 s
GREEN	Signal processing	Skin boxes	Verkruysse et al. 2008: inverted green channel (haemoglobin absorbs green). No rejection of motion or lighting; a lower bound.	Every 2 s
TS-CAN	Deep learning	72 x 72 face crops, 30 Hz	Liu et al. 2020: two-branch temporal-shift convolutional attention network (appearance + motion). rPPG-Toolbox weights trained on PURE.	Every 12 s, Apple GPU (MPS), about 2 s
PhysNet	Deep learning	Same crops	Yu et al. 2019: 3-D convolutional encoder-decoder over space-time. Runs on the CPU on Macs because MPS lacks its 3-D layers.	Every 12 s, CPU, about 2-5 s
EfficientPhys	Deep learning	Same crops	Liu et al. 2023: single-branch network with built-in normalisation, no separate preprocessing.	Every 12 s, MPS, about 1 s
Microphone (PCG)	Acoustic	INMP441 node audio	Heart-sound envelope autocorrelation; needs the mic on the chest.	Continuous

The deep models are pretrained on PURE and never trained on your face or on the test datasets (cross-dataset). The spread and median across all rows are shown under the chart.

6.3 Camera modes

	Mac webcam	ESP32-S3 + GC2145 node
Frame rate	About 30 fps	Skin colour sums at 10 fps; video 5 fps (JPEG q65)
Where colour is measured	Host computer	On the node (raw pixels, clipped and dark pixels excluded); host face gate decides which samples count
Exposure	Webcam automatic	Skin-metered exposure control (registers P0:0x03/0x04)
Deep models	Yes (>= 12 fps)	No (video too slow)
Microphone	Node mic if the node is connected	Node mic

7. Benchmark: evaluation modes and results (UBFC-rPPG)

The same code is evaluated offline on public datasets (Colab notebook, *benchmark/*). Reference: the contact PPG recorded with each video, processed exactly like the predictions. Windows of 30 s, step 10 s, plus whole-video estimates. Metrics: MAE, RMSE, MAPE, Pearson r, share within 5 BPM, coverage (share of windows with an output) and processing time per 30 s window.

Evaluation mode	Meaning
SonicVitals (raw)	Single-window estimate in every window, ignoring the gate.
SonicVitals (gated)	Only windows that pass the gate; coverage reports how many.
SonicVitals (display)	Replays the dashboard rules (2 s updates, confirmation, median, EMA, hold).
SonicVitals (tracked) / (tracked, high conf.)	Ensemble spectrum + Bayes tracker, all windows / only confidence >= 0.8. Added after the 15 Oct run; numbers pending the next Colab run.
GREEN, CHROM, POS	Standard baselines on the whole face box (and, as an ablation, on the SonicVitals skin boxes).
TS-CAN, PhysNet, EfficientPhys [train -> test]	Pretrained rPPG-Toolbox weights, tested only on a dataset they were not trained on.

Results: UBFC-rPPG, first 10 subjects, 30 s windows (Colab T4, 15 Oct 2026)

Method	Coverage	MAE	RMSE	r	Within 5 BPM	ms / 30 s
GREEN	100 %	22.04	34.21	0.07	47 %	2.4
CHROM	100 %	2.40	6.48	0.93	86 %	19.6
POS	100 %	2.08	5.41	0.95	86 %	78.6
SonicVitals (raw)	100 %	2.30	4.92	0.96	89 %	71.8
SonicVitals (gated)	81 %	**1.56**	**2.73**	**0.99**	**93 %**	71.8
SonicVitals (display)	86 %	2.45	3.30	0.98	87 %	71.8
TS-CAN [PURE->UBFC]	100 %	1.68	4.95	0.96	89 %	278.5
PhysNet [PURE->UBFC]	100 %	1.90	5.00	0.95	86 %	159.9
EfficientPhys [PURE->UBFC]	100 %	3.88	9.09	0.88	81 %	141.9

MAE/RMSE in BPM, over covered windows only. The gated mode trades 19 % of windows for the lowest error; errors of raw and deep models include their difficult windows. PURE and VIPL-HR runs are still to be done.

8. Formulas

Quantity	Formula
CHROM pulse	Per 3 s window, Rn, Gn, Bn = channel / window mean - 1; X = 3Rn - 2Gn; Y = 1.5Rn + Gn - 1.5Bn; s = X - (std X / std Y) Y; windows overlap-added.
POS pulse	Per 1.6 s window, C = channel / mean; S1 = G - B; S2 = -2R + G + B; p = S1 + (std S1 / std S2) S2; overlap-added.
de Haan SNR	10 log10 [P(|f - f0| <= 0.1 Hz or |f - 2f0| <= 0.2 Hz) / P(rest of 0.5-4 Hz)]
Spectral concentration (SQI)	P(|f - f0| <= 0.1 Hz) / P(0.75-2.5 Hz)
Illumination rectification	beta_c = / on 0.6-3 Hz relative traces, clipped to 0-1.5; skin_c -= beta_c bg_c (skipped if bg RMS < 0.3 x skin RMS).
Artifact repair	d = diff(log RGB); frames with |d - median| > 4 x 1.4826 MAD set to 0; re-integrate.
Bayes tracker	prior = posterior convolved with Gaussian (SD 3 BPM per 2 s), mixed 98/2 with uniform; posterior = prior x spectrum^1.5, normalised; HR = argmax; confidence = mass within +-3 BPM.
PCG HR	Shannon energy E = -x^2 log x^2 (20 ms smoothing); HR = 60 / lag of the autocorrelation peak in 0.4-1.33 s.
RMSSD	sqrt(mean(diff(IBI)^2)) over >= 20 clean inter-beat intervals (0.4-1.5 s, within 20 % of median).
Fusion	If |HR_o - HR_a| <= 8: (w_o HR_o + w_a HR_a)/(w_o + w_a), w_o = SQI x gate, w_a = rho; else the higher-weight channel.
SpO2 ratio	R = (AC_red / DC_red) / (AC_blue / DC_blue); SpO2 = A - B R only after calibration.

9. Limits to keep in mind

• The camera pulse is about 0.3 % of skin brightness. Head-pose shading and webcam auto-exposure produce changes of several %; the chain removes much of this, but in a still, well-lit face SNR can still fall below the gate. The amber tracked value is a best guess, not a measurement.

• Live webcam values have no reference unless you count your pulse or use a finger oximeter at the same time. Accuracy claims rest on the UBFC benchmark only.

• SpO2 and the Diagnostics Studio indices are uncalibrated and not clinical.

• The microphone channel needs the INMP441 on the chest; held in the air it gives no heart rate.